Online Appendix for
"New Area- and Population-based Geographic Crosswalks for U.S. Counties and Congressional Districts, 1790–2020"*

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*The area- and population-based crosswalks produced in the paper, teaching material, as well as code and data for the replication exercise can be downloaded at https://doi.org/10.3886/E150101.

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1 Overview of Data

1.1 Data Used to Produce Crosswalks

Our crosswalks are based in part on the U.S. county shapefiles from https://www.nhgis.org/. For the period 1790 to 2000, we use the shapefiles based on the 2000 TIGER/Line county boundary definitions. For 2010 and 2020, these are not available. We therefore use contemporaneous TIGER/Line shapefiles for those years instead. The shapefiles for 2010 and 2020 are sourced from https://www.census.gov/.

Shapefiles for congressional districts (CD) are from Lewis, DeVine, Pritcher and Martis (2021) for the 1st to 114th Congress, as available at https://cdmaps.polisci.ucla.edu/. The shapefiles for CDs from the 115th and 116th Congress are obtained from https://catalog.data.gov/dataset/tiger-line-shapefile-2016-nation-u-s-115th-congressional-district-national.

Historical population distribution maps used to construct M2–M4 come from Fang and Jawitz (2018). These are described in greater detail in the next section. We use U.S. Census Bureau tract shapefiles and population data for 1960 to construct a population distribution raster for that year, both from NHGIS, since Fang and Jawitz (2018) do not have urban population data for that year.

Historical property records maps used to construct M5–M6 come are based on Leyk et al. (2020) and available for download at Uhl and Leyk (2020a) and Uhl and Leyk (2020b). These are described in greater detail in Section 3.

1.2 Data Used in Replication Exercise

For the replication of Lee et al. (2004) in the application section of the main paper, we use the data and code to replicate Table 2 from Enrico Moretti's website. The data can be accessed via https://eml.berkeley.edu/~moretti/data3.html. Their replication files include the data from the U.S. Census extracts for 1960-90. To reconstruct CD level from county-level census data, we use data from the U.S. Census of Population and Housing for 1960 and 1970 from Haines (2010), as well as with information from the County and City Data Books for the years 1962, 1967, 1972 1977, 1983, and 1994. This allows us to test the performance of different crosswalks (area- versus population-based) when crosswalking the county-level data to the CD-level, in comparison to the official CD-level data produced by the Census Bureau. Urban population data for 1990 come from the 1990 U.S. Census, which is separately sourced from NHGIS.

2 Overview of Spatial Models in Fang and Jawitz (2018)

2.1 Data

To estimate spatial models of population distribution for the conterminous U.S., Fang and Jawitz (2018) use the following data:

1. Total and urban (threshold 2,500+) population at the county level from the National Historical Geographic Information System (https://www.nhgis.org/) between 1790 and 2010 (excluding 1960, as these data are missing, and 2020, as these are not yet available),

2. Data on water bodies from the National Hydrography Dataset (http://nhd.usgs.gov/) including lakes, ponds, marsh and swamp land, and other minor water bodies as of 2001,

3. Boundaries of protected areas from the National Gap Analysis Program (http://gapanalysis.usgs.gov/padus/) as of 2012,

4. Elevation data from the NASA Shuttle Radar Topography Mission Version 3.0 (http://www2.jpl.nasa.gov/srtm/) as of 2013,

for 7,754,146 square-kilometer grid cells in the conterminous United States. They caution that protected areas were established in more recent times and that areas settled by American Natives will undercount population as this group was only fully included in the U.S. censuses from 1900 onward. An illustration of the spatial variation of their data for the year 2000 is shown in Figure A2.

Defining the Spatial Extent of Urban Areas

Fang and Jawitz (2018) measure the areal extent of 3,610 urban areas using the 2000 U.S. Census and project area backward in time using a power law relationship between an urban area's population and its spatial extent,

$$A_{U,\varphi} = \alpha_\delta P_{U,\varphi}^{\beta_\delta} \tag{1}$$

where $A_{U,\varphi}$ is the spatial extent of urban area and urban areas are indexed by φ in U.S. Census Bureau division δ, α_δ and β_δ are the coefficients of the power function, which are assumed to be constant over time,[1] and $P_{U,\varphi}$ is the population size. Using the log-transformed version of the model and historical population data from the census, they then estimate the historical area size of the urban areas in their sample.

Defining Inhabitable Areas

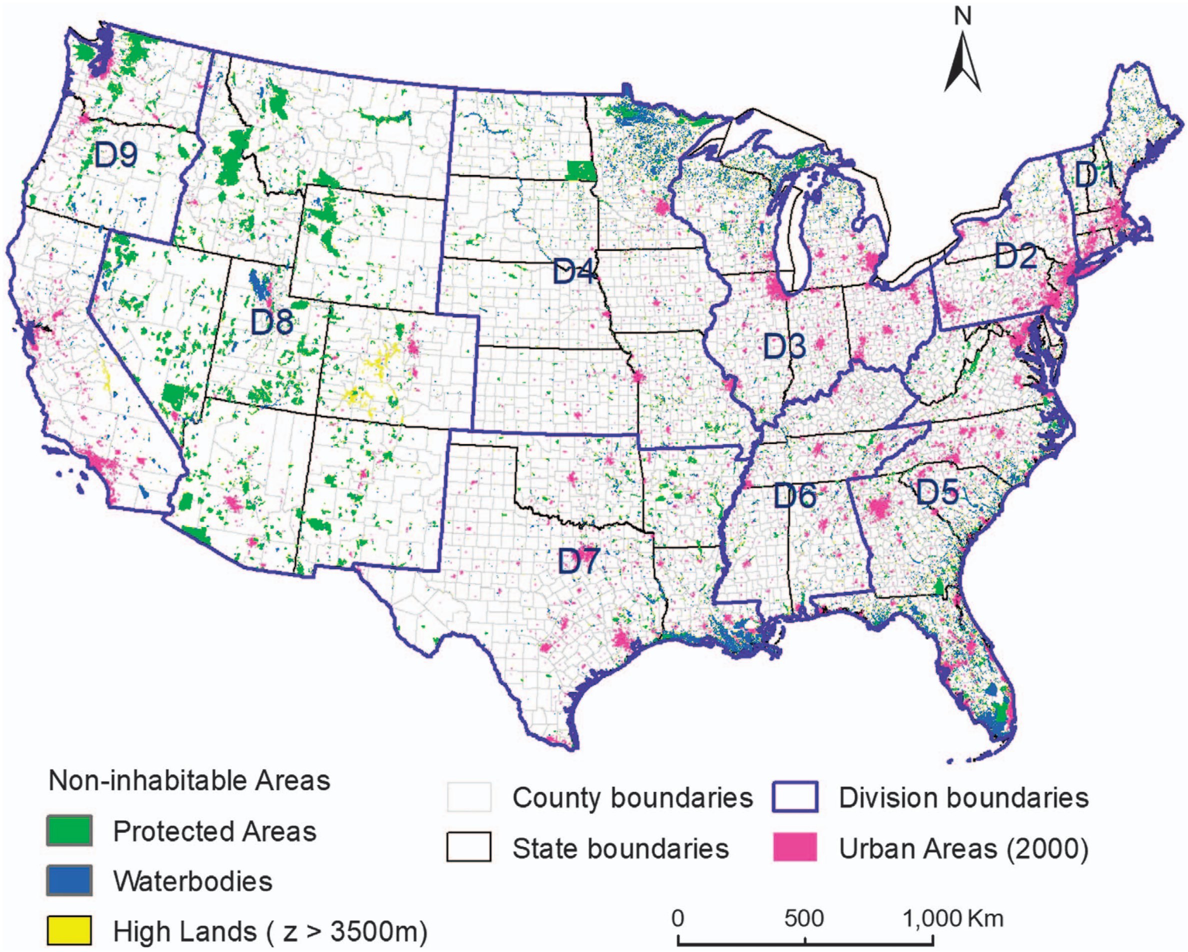
Fang and Jawitz (2018) define inhabitable areas as those that are not water bodies larger than 1 square kilometer, protected areas, or areas with an elevation of more than 3,500 meters.

2.2 Models

Fang and Jawitz (2018) employ five spatial models of population distribution for the conterminous U.S. The first model corresponds to county-level population distributions. This assumes a

[1] They motivate this assumption referencing other historical models of urbanization and urban extent in England, China, Pre-Hispanic Mexico, and Japan.

Figure A1: Geographic Distribution of Data Features Used by Fang and Jawitz (2018) for the Year 2000



Note: Map showing variation in the main data sources used by Fang and Jawitz (2018) for the year 2000. D1-D9 refer to U.S. Census Bureau divisions. Source: Figure 1 in (Fang and Jawitz, 2018, p. 3).

uniform population distribution within counties, with each one-by-one kilometer grid cells having the same population value within a given county. The second model differentiates between rural and urban areas, using the urban areal extents and urban population stock data described above. Within a county, urban population is distributed uniformly within urban areas and the remaining non-urban population is distributed uniformly within rural areas. The third model also does this but first excludes non-inhabitable areas, as described above. The fourth model extends the third by also multiplying the population raster by topographic suitability weights. These four models correspond to our M1–M4, respectively. We do not discuss or utilize their fifth model, which incorporates information from a constructed "socioeconomic desirability" index. For most applications in the social sciences this allocation of population is problematic as it is based on potentially endogenous variables, such as the distance from the periphery to the core of cities.[2]

Alternative Models for 1960

Because digital urban population data are missing for the 1960 U.S. Census, Fang and Jawitz (2018) do not develop models for that year. In lieu of this, and in the interest of completeness, we still construct crosswalks based on the area-based M1 for 1960, assuming a uniform population distribution within counties. We also construct crosswalks based on an alternative M2, in which we use the most granular population data possible: (i) population and boundary data at the 1960 U.S. Census tract level, where available (coverage is most urban areas), and (ii) population and boundary data at the 1960 county level otherwise. This is akin to adopting a uniformity assumption within rural counties, i.e., where population distribution is likely to be relatively homogeneous anyway. These population and boundary data are transformed into a raster with square kilometer grid cells, to match the Fang and Jawitz (2018) M2 for other years. In the crosswalk file, these weights take the places of M2–M4 for 1960.

Modeling Population Distribution for 2020

Given that Fang and Jawitz (2018) predates the 2020 U.S. Census, and urban or census-tract-level population data are not yet available, we utilize urban population data from 2010 and the corresponding models (M2–M4) from Fang and Jawitz (2018) in order to model population distributions within counties for 2020 counties that are being harmonized to CDs or to other counties.

3 Overview of Spatial Models in Leyk et al. (2020)

To construct maps of historical settlements for the conterminous U.S., Leyk et al. (2020) use settlement layers from the Historical Settlement Data Compilation for the United States (HISDAC-US), which is itself derived from historical property records found in the Zillow Transaction and Assessment Database (ZTRAX). These settlement layers are granular, based on 250×250 meter grid cells, and built for 5 year periods beginning in 1810. The database includes a variety of measures:

- The historical "built-up areas" measure (BUA) defines for each grid cell the presence of at least one built-up structure in a given year (built-up$= 1$).
- The historical "built-up property records" measure (BUPR) defines for each grid cell the number of built-up property records in a given year.
- The historical "built-up property locations" measure (BUPL) defines for each grid cell the number of unique locations of built-up property records in a given year (i.e., independent of ownership). This is highly correlated with the BUPR measure.

[2] Unlike geographic and topographic features, distance based on a gravity model is also likely to change the most over time due to changes in available transportation methods.

- The “first built-up year” measure (FBUY) defines for each grid cell the earliest construction year on record.

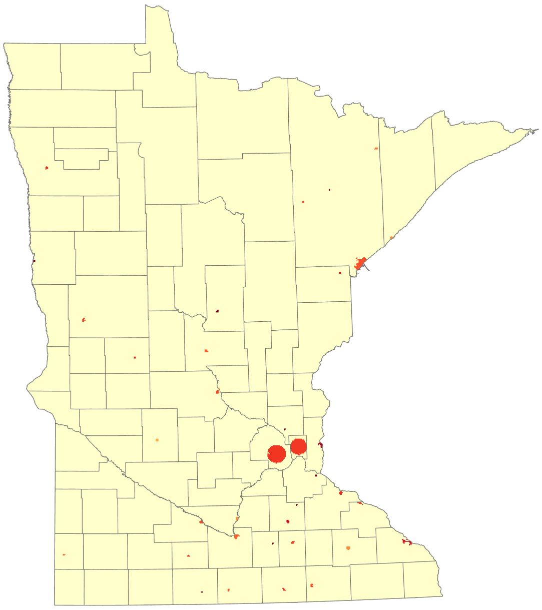
The first two measures become correspond to our models M5 and M6, respectively, for all census years 1810 through 2020. As with M1–M4, we use the 2010 map to construct our models for 2020.

Built-up property records are importantly highly correlated with population size, as the authors show in Leyk et al. (2020). Comparisons with county-level population data for 1860–2010 in their Table 1 show that a one unit increase in built-up property records within a county is associated on average with 2.68 (0.01) additional residents, with these records accounting for nearly 93% of the variation in total population size over time across sample counties. On that basis, property records provide an accurate and granular proxy for historical population counts.

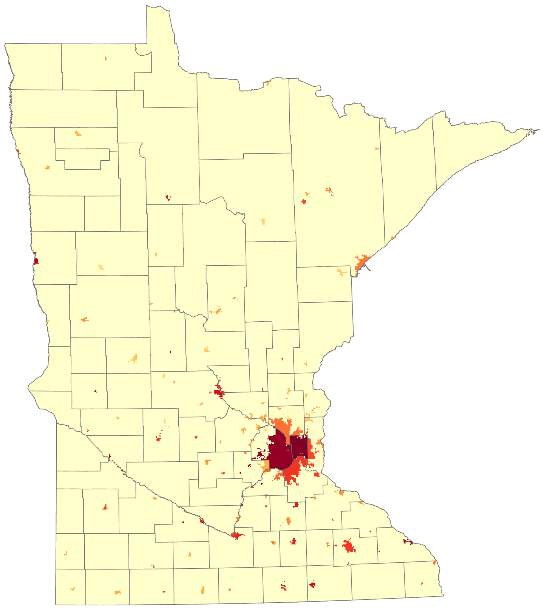
Figures A2 and A3 provide different visualizations of the weights to construct our crosswalks in our example of Minnesota. Figure A2 shows the population-based weights for the M2 (panels a and b) and M4 (panels c and d) models in 1900 and 2000, respectively. Figure A3 maps the spatial extent of built-up areas (panels a and b) and the built-up property records (panels c and d) for the same state and time periods.

Figure A2: Examples of M2 and M4 from Fang and Jawitz (2018), Used to Construct Our M2 and M4 Weights

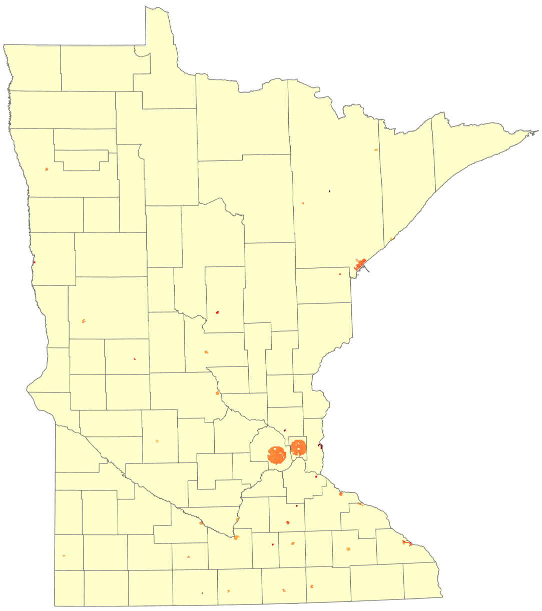
(a) M2, Minnesota, 1900



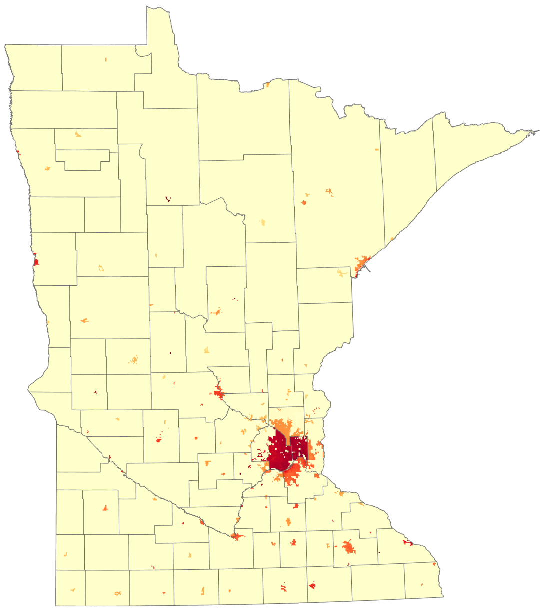
(b) M2, Minnesota, 2000



(c) M4, Minnesota, 1900

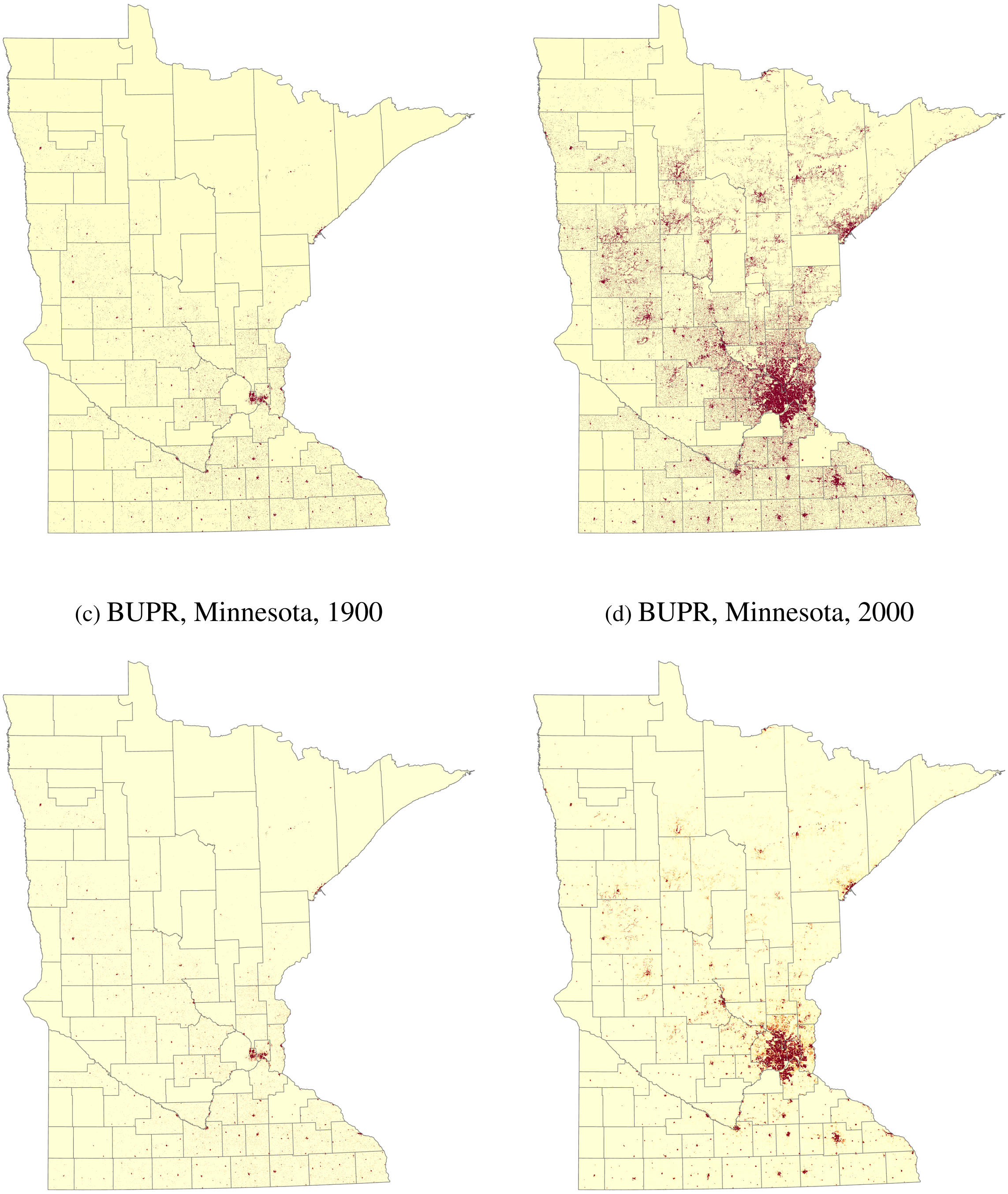


(d) M4, Minnesota, 2000



Note: This figure shows the land area of the state of Minnesota with built-up distribution information for 1900 and 2000, where darker orange a greater number of residents per square kilometer for M2 and M4 (used to construct our weights of the same names). The gray boundaries show the state's county boundaries as of the 2010 U.S. Census. County shapefiles are from Manson, Schroeder, Van Riper, Kugler and Ruggles (2020). Population distribution estimates for 1900 and 2000 are from Fang and Jawitz (2018).

Figure A3: Examples of Built-Up Areas (BUA) and Built-Up Property Records (BUPR) from Leyk et al. (2020), Used to Construct Our M5 and M6 Weights


(a) BUA, Minnesota, 1900 (b) BUA, Minnesota, 2000

(c) BUPR, Minnesota, 1900 (d) BUPR, Minnesota, 2000

Note: This figure shows the land area of the state of Minnesota with built-up distribution information for 1900 and 2000, where darker orange implies at least one build-up property within the grid cell by the year for BUA (used to construct our M5), or a greater number of individual built-up property record by the year for BUPR (used to construct our M6). County shapefiles are from Manson et al. (2020). Built-up property distribution information for 1900 and 2000 are from Leyk et al. (2020).

4 Construction of the Geographic Crosswalks

This paper makes a number of contributions to the study of spatial phenomena in the social sciences, with a wide range of applications for urban economists, political scientists, and economic historians, among others. Most notably, we extend existing area-based approaches to harmonizing county boundaries across U.S. censuses over time, by relaxing standard uniformity assumptions regarding population distribution within counties for the contiguous U.S. (i.e., excluding Alaska and Hawaii). To do this, we rely on maps based on historical population estimates for 1×1 kilometer grid cells from Fang and Jawitz (2018) (for M2–M4) and maps based on historical property records for 250×250 meter grid cells from Leyk et al. (2020) (for M5–M6). These capture sub-county variation in the population distribution for counties being harmonized, to the extent that more or less of its population may be located in some "parts" relative to others. Given that harmonization frequently involves spatial disaggregation of counties prior to the re-aggregation to the boundaries of some "reference" unit, this better ensures that initial county stock variables will be properly weighted in the disaggregation process prior to being re-aggregated to different reference county boundaries.

We then apply this innovation in order to construct wholly original county-to-congressional district (CD) crosswalks, spanning the entirety of U.S. congressional history, from 1790 to 2020. These crosswalks can be used to aggregate county-level data to the CD-level. Because CD boundaries often do not align with county boundaries, county data must often be spatially disaggregated before being re-aggregated to the CD level. This renders the models of historical sub-county population distribution from Fang and Jawitz (2018) and Leyk et al. (2020) quite important.

We now describe this process of disaggregation and re-aggregation, used to generate our county-to-county and county-to-CD crosswalks, beginning with the area-based crosswalks used previously in the literature.

4.1 Area-based Crosswalks

Area-based harmonization procedures entail a simple process of spatial disaggregation and re-aggregation. To construct our county-to-CD crosswalks, this involves intersecting a county map from a particular census year with a CD map from a particular Congress year. Counties are then disaggregated into a set of sub-county units ("county-parts"), based the CD in which they are located. We then calculate the areas (in square meters) of all counties, all CDs, and all county-parts, based on a "USA Contiguous Albers Equal Area Conic" projection. Once counties are disaggregated based on CD intersections, county-parts are re-aggregated based on their CD, with the sum of the areas of the county-parts matching the area of the whole CD. This is the same process outlined in Eckert, Gvirtz, Liang and Peters (2020) and used to construct the county-to-county crosswalks featured in Hornbeck (2010), Perlman (2021), and Bazzi, Fiszbein and Gebresilasse (2020), among others.

How are the various data values of the initial counties (e.g., total population, total number of Blacks) associated with CDs in this process? Under an area-based procedure, each county-part is assigned each of its county's data values, weighted by the share of the county's total *area* that belongs to that county-part. These weights add up to 1 for each county. A given CD's data values are in turn the aggregates of these weighted values, summed across all counties that have a county-part located in that CD. For our area-based weights to be appropriate in settings where county and CD boundaries overlap, the following condition and proposition are relevant:

Proof of Proposition

Proposition 1. *Suppose all counties satisfy:*

Assumption (Uniformity)**.** *Let C be any continuous, two-dimensional county with area $c > 0$ and a vector of positive and finite values $P = (p_1, p_2, ..., p_n)$. Let A be any continuous, two-dimensional subset of C with area $ac \in (0, c)$ and a vector of positive and finite values $R = (r_1, r_2, ..., r_n)$. C satisfies uniformity in population distribution if $R = aP$ for all $A \subset C$.*

Then our area-based crosswalk will accurately map county-level values to the congressional district level for all districts.

Proof. Let D be any continuous, two-dimensional congressional district with area $d > 0$ and a vector of positive and finite values $Q = (q_1, q_2, ..., q_n)$. Suppose D can be decomposed into 1 county-part each from M counties $j = 1, ..., m$, each with finite area $a_j c_j$ and a vector of positive and finite values $R_j = (r_{j1}, r_{j2}, ..., r_{jn})$, such that:

$$\sum_{j=1}^{M} a_j c_j = d,$$

$$\sum_{j=1}^{M} R_j = Q,$$

where a_j is the share of county j's area belonging to its county-part that lies in D. As such, $(a_1, a_2, ..., a_m)$ is the vector of weights associated with our area-based crosswalk, which map values from counties $j = 1, ..., m$ to D.

Yet suppose in actuality that our area-based crosswalk does not accurately map county-level values to D, such that:

$$Q \neq \sum_{j=1}^{M} a_j P_j,$$

where P_j is the vector of positive and finite values associated with county j. It follows that $a_j P_j \neq R_j$ for at least one county j, a violation of uniformity. $\square$

4.2 Population-based Crosswalks

We also construct population-based crosswalks, which rely on a relaxation of the uniformity assumption. To do this, we use information on historical sub-county population distribution by Fang and Jawitz (2018) and Leyk et al. (2020). The former provide historical population counts for 1×1 kilometer grid cells, which we use to construct a set of population-based weights. These include: (i) model 2 (M2), which is based on a division of counties into urban and rural areas, with urban population counts being distributed around city centers according to a power law scaling relationship; (ii) model 3 (M3), which is is based on a version of M2 that first excludes non-inhabitable areas, such as bodies of water; and (iii) model 4 (M4), which is based on a version of M3 that also weights population counts based on topographic suitability. In contrast, Leyk et al. (2020) derive proxies for historical population size for 250×250 meter grid cells based on historical property records data, which they show to be highly correlated with local population size. We use these to construct two further models: (i) model 5 (M5), which is based on their binary measure of "built-up area," which assigns a value of 1 to a grid cell if it contains at least one built-up property record in a given year, and (ii) model 6 (M6), which is based on the "built-up property" counts themselves, summing the number of records (e.g., building units) within the grid cell in a given year. We describe these data and models in detail above.

To relax the uniformity assumption, we no longer base the disaggregation of county-level data on relative area but rather relative population size. These various population distribution raster maps let us calculate the total population counts (or property-based proxy) of each county polygon for each census year, as well as approximate counts for each county-part polygon within a county that lies in a different CD. As with our area-based crosswalk, the ratio of these counts in the county-part relative to the entire "origin" county provides a weight with which to multiply a county's stock data prior to its aggregation to the CD level.

4.3 Using ArcMap and Stata to Construct Crosswalks

Although our crosswalks currently cover the entirety of U.S. congressional history and of the U.S. censuses, time will render them incomplete. We are therefore describing the data and steps taken to (i) generate necessary spatial data and (ii) construct crosswalk weights for a given county-CD pair. After providing these examples, we will discuss step-by-step how to apply these weights and harmonize county boundaries to those of a contemporaneous CD, including relevant samples of Stata code.

4.3.1 Guide for Generating Spatial GIS Data

We use ArcMap to generate the necessary spatial information for crosswalks, based on the data sources described in the first part of this Online Appendix. This process goes as follows for each county-CD crosswalk:

We first calculate the area (in square meters) of each county polygon within each county shapefile. Working within a NAD 1983 data frame, all area calculations are based on a "USA Contiguous Albers Equal Area Conic" projection. We then intersect the county shapefile with a given CD polygon shapefile, using the "intersect" tool. The output table from this intersection provides the full set of county-parts for each county.

For area-based crosswalks, we calculate the area (in square meters) of each county-part. For population-based crosswalks, we use the "zonal statistics table" tool to calculate the sum of population (or property-based proxy) counts within each county-part, based on the raster grid cells for a given model M2–M6 in a given census year. These produce five separate tables, one each for M2–M6, with the same county-part identifiers used in the intersection output table. We then export all six tables to CSV.

In Stata, we import each CSV separately. We then use the identifiers for the county-parts from ArcMap to merge the M2, M3, M4, M5, M6 and intersection output tables. Total population (or property-based proxy) counts for each county for M2–M6 are calculated by summing said counts of all county-parts within each given county.

To generate weights for area-based crosswalks, we calculate the ratio of county-part-to-origin-county area for each county-part. To generate weights for each population-based crosswalks, population (or property-based proxy) counts are divided by the total counts for the origin county for each population-based model. Miscellaneous minor edits are made to make sure county and CD identifiers are consistent across crosswalk years and to fix a few errors relating to duplicate county-parts (e.g., in the case of pre-1960s CDs in states with both at-large and within-state CDs overlapping each other).

5 Step-by-step Guide for Applying the Crosswalks in Stata

Below we provide a simple example using Stata to show step-by-step how to crosswalk county level aggregates to congressional district boundaries. Here we wish to crosswalk total population and Black population (in levels) from 1960 to the boundaries of the 88th Congress.

First, take total population and Black population at the county level and prepare the data for merging with the crosswalk file.

```
. use state county level statefip counfip var3 using "1962_cnty_and_city_data_book.dta"
(Historical, Demographic, Economic, and Social Data: The United States, 1790-2002)

.
. gen census = 1960

.
. * merge with county level data on Black population
. merge 1:1 state county using "1960_census_county.dta", keepusing(negmtot negftot)

    Result                           # of obs.
    -----------------------------------------
    not matched                             2
        from master                         2  (_merge==1)
        from using                          0  (_merge==2)

    matched                             3,183  (_merge==3)
    -----------------------------------------
```

```
. drop _merge

.
. * keep counties only (level 1) and drop D.C. (no congressional representation)
. keep if level==1 & state!=98
(53 observations deleted)

.
. * generate the variables of interest in levels
. gen black = negmtot + negftot

. gen totpop = var3

.
. * keep only relevant variables for the cross walk
. keep state county census black totpop

.
. * rename county identifiers for merging with the crosswalk
. ren (state county) (icpsrst icpsrcty)
```

Second, use the county and state identifiers to merge the crosswalk file to the data in a 1:m merge. This expands the set of counties to the full set of county-parts belonging to each congressional district intersecting a given county.

```
. * Merge with crosswalks
. merge 1:m icpsrst icpsrcty using "Crosswalk_1960_88.dta"

    Result                           # of obs.
    -----------------------------------------
    not matched                             2
        from master                         1  (_merge==1)
        from using                          1  (_merge==2)

    matched                             7,368  (_merge==3)
    -----------------------------------------

. drop _merge
```

Third, multiply each stock variable with the relevant weights, using the weights from all six models and keeping track of county-parts with missing data or undefined weights.

```
. * Weight county count data by weights
. qui ds black totpop

. foreach v in `r(varlist)´ {
. * If any one "county part" has a missing value, may want to mark the whole of the
. * aggregated district to have a missing value as well, especially if that county
. * part makes up a sizable part of the district
  1.         replace `v´ = -999999999999999 if(`v´==.) & cnty_part_area/cd_area>0.05
.         *Apply weights (for all 4 models)
  2.         gen `v´_m1 = `v´*m1_weight
  3.         gen `v´_m2 = `v´*m2_weight
  4.         gen `v´_m3 = `v´*m3_weight
  5.         gen `v´_m4 = `v´*m4_weight
  6.         gen `v´_m5 = `v´*m5_weight
  7.         gen `v´_m6 = `v´*m6_weight
. }
(0 real changes made)
(2 missing values generated)
(36 missing values generated)
```

```
(36 missing values generated)
(36 missing values generated)
(642 missing values generated)
(642 missing values generated)
(0 real changes made)
(2 missing values generated)
(36 missing values generated)
(36 missing values generated)
(36 missing values generated)
(642 missing values generated)
(642 missing values generated)

.
. qui ds black* totpop*

. foreach v in `r(varlist)´ {

. * If any one "county part" has a missing value due to a missing weight,
. * may want to mark the whole of the aggregated district to have a missing
. * value as well
.         replace `v´ = -999999999999999 if(`v´==.) & cnty_part_area/cd_area>0.05
. }
(0 real changes made)
(1 real change made)
(12 real changes made)
(12 real changes made)
(12 real changes made)
(167 real changes made)
(167 real changes made)
(0 real changes made)
(1 real change made)
(12 real changes made)
(12 real changes made)
(12 real changes made)
(167 real changes made)
(167 real changes made)
```

Fourth, collapse the weighted data on the CD identifier. The unit of observation is now the congressional district.

```
. * Collapse by congressional district
. collapse (sum) black_m* totpop_m*, by(census congress cd_state cd_statefip ///
>                                              cd_stateicp district id cd_area)

.
. * Correct districts with missing "county parts" and otherwise round to nearest integer
. qui ds black_m* totpop_m*

. foreach v in `r(varlist)´ {
  2.
.         replace `v´ = . if(`v´<=0)
  3.         replace `v´ = round(`v´)
  4.
. }
(1 real change made, 1 to missing)
(387 real changes made)
(3 real changes made, 3 to missing)
(160 real changes made)
(3 real changes made, 3 to missing)
(160 real changes made)
(3 real changes made, 3 to missing)
(160 real changes made)
(3 real changes made, 3 to missing)
(150 real changes made)
```

```
(3 real changes made, 3 to missing)
(150 real changes made)
(1 real change made, 1 to missing)
(383 real changes made)
(3 real changes made, 3 to missing)
(159 real changes made)
(3 real changes made, 3 to missing)
(159 real changes made)
(3 real changes made, 3 to missing)
(159 real changes made)
(3 real changes made, 3 to missing)
(150 real changes made)
(3 real changes made, 3 to missing)
(144 real changes made)
```

The final step also rounds the relevant variables or replaces them as missing in cells where this is appropriate.

6 Tables from Our Replication of Lee et al. (2004)

Table A1: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data

	Difference in District Population Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
Total pop (M1)	-92,262.9***	-72,968.3***	-23,473.9**	-24,417.8*	-34,212.2	-12,495.7
	(12,117.1)	(12,874.6)	(11,741.6)	(12,977.4)	(22,510.6)	(21,968.6)
Total pop (M2)	-37,081.3***	-18,546.0***	-3,286.6	-3,727.6	-1,255.2	-336.5
	(5,975.7)	(5,935.8)	(6,254.6)	(7,972.6)	(12,973.8)	(13,872.5)
Total pop (M3)	-38,286.8***	-19,051.5***	-3,706.1	-4,059.8	-1,240.9	13.9
	(6,224.5)	(6,156.6)	(6,348.2)	(8,065.7)	(13,139.5)	(14,200.1)
Total pop (M4)	-32,030.1***	-14,839.0**	-2,192.7	-3,198.2	1,262.0	3,320.0
	(5,950.8)	(5,905.5)	(5,958.1)	(7,710.2)	(12,850.0)	(13,732.5)
Total pop (M5)	-64,413.7***	-47,177.6***	-17,042.3**	-13,148.8	-17,519.3	-4,635.7
	(7,113.2)	(7,351.6)	(7,845.1)	(9,414.8)	(15,465.8)	(15,524.0)
Total pop (M6)	-20,360.6***	-7,620.1*	-2,138.8	-2,258.4	2,145.1	7,709.1
	(4,137.0)	(4,215.2)	(5,507.0)	(7,500.8)	(13,164.5)	(11,929.9)
Total pop (LMB)	-1,817.6	3,019.9	4,961.5	3,211.1	8,640.6	2,008.0
	(3,517.3)	(3,723.4)	(4,562.7)	(5,524.2)	(8,427.0)	(9,258.1)
Bandwidth	All	+/− 25	+/− 10	+/− 5	+/− 2	Polynomial
Observations	13,231	10,065	4,086	2,030	794	13,211

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. The unit of observation is the district-congress. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data (I)

	Difference in District Income Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
Log income (M1)	-0.039***	-0.002	0.026*	0.033*	0.027	0.007
	(0.013)	(0.013)	(0.015)	(0.018)	(0.026)	(0.028)
Log income (M2)	-0.038***	-0.001	0.026*	0.033*	0.027	0.008
	(0.013)	(0.013)	(0.014)	(0.018)	(0.026)	(0.028)
Log income (M3)	-0.039***	-0.001	0.026*	0.033*	0.027	0.008
	(0.013)	(0.013)	(0.015)	(0.018)	(0.026)	(0.028)
Log income (M4)	-0.039***	-0.001	0.026*	0.033*	0.026	0.008
	(0.013)	(0.013)	(0.015)	(0.018)	(0.026)	(0.028)
Log income (M5)	-0.039***	-0.006	0.026*	0.037*	0.018	0.023
	(0.014)	(0.013)	(0.015)	(0.019)	(0.027)	(0.029)
Log income (M6)	-0.038***	-0.005	0.026*	0.038**	0.019	0.024
	(0.014)	(0.013)	(0.015)	(0.019)	(0.027)	(0.029)
Log income (LMB)	-0.087***	-0.037***	0.014	0.027	0.031	0.053*
	(0.013)	(0.013)	(0.014)	(0.018)	(0.026)	(0.029)
Bandwidth	All	+/− 25	+/− 10	+/− 5	+/− 2	Polynomial
Observations	13413	10229	4174	2072	810	13393

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Standard errors are in parenthesis. The unit of observation is the district-congress. Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data (II)

	Difference in District Urban Pop. Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
% Urban (M1)	0.050***	0.052***	0.045***	0.045***	0.055***	0.042*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.022)
% Urban (M2)	0.052***	0.054***	0.045***	0.045***	0.055***	0.043*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.023)
% Urban (M3)	0.051***	0.053***	0.045***	0.045***	0.055***	0.042*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.023)
% Urban (M4)	0.052***	0.054***	0.045***	0.045***	0.055***	0.043*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.023)
% Urban (M5)	0.048***	0.047***	0.037***	0.039***	0.046**	0.042*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.023)
% Urban (M6)	0.050***	0.049***	0.038***	0.041***	0.046**	0.044*
	(0.011)	(0.011)	(0.012)	(0.014)	(0.021)	(0.023)
% Urban (LMB)	0.070***	0.066***	0.054***	0.054***	0.056**	0.053**
	(0.011)	(0.011)	(0.013)	(0.015)	(0.023)	(0.025)
Bandwidth	All	+/− 25	+/− 10	+/− 5	+/− 2	Polynomial
Observations	13413	10229	4174	2072	810	13393

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Standard errors are in parenthesis. The unit of observation is the district-congress. Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data (III)

	Difference in District Blacks Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
% Black (M1)	0.061***	0.037***	0.013***	0.007	-0.001	-0.006
	(0.005)	(0.004)	(0.005)	(0.006)	(0.009)	(0.010)
% Black (M2)	0.062***	0.037***	0.013***	0.007	-0.001	-0.005
	(0.005)	(0.004)	(0.005)	(0.006)	(0.009)	(0.010)
% Black (M3)	0.062***	0.037***	0.013***	0.007	-0.001	-0.006
	(0.005)	(0.004)	(0.005)	(0.006)	(0.009)	(0.010)
% Black (M4)	0.062***	0.038***	0.013***	0.007	-0.001	-0.006
	(0.005)	(0.004)	(0.005)	(0.006)	(0.009)	(0.010)
% Black (M5)	0.056***	0.033***	0.007	0.001	-0.006	-0.012
	(0.005)	(0.005)	(0.005)	(0.006)	(0.010)	(0.011)
% Black (M6)	0.057***	0.034***	0.008	0.001	-0.005	-0.012
	(0.005)	(0.004)	(0.005)	(0.006)	(0.010)	(0.011)
% Black (LMB)	0.083***	0.043***	0.014***	0.003	-0.003	-0.053***
	(0.006)	(0.005)	(0.005)	(0.006)	(0.009)	(0.012)
Bandwidth	All	+/− 25	+/− 10	+/− 5	+/− 2	Polynomial
Observations	13413	10229	4174	2072	810	13393

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Standard errors are in parenthesis. The unit of observation is the district-congress. Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data (IV)

	Difference in District Manufacturing Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
% Manufacturing (M1)	-0.003	-0.000	0.004*	0.005*	0.004	0.002
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (M2)	-0.002	0.000	0.004*	0.005*	0.004	0.002
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (M3)	-0.002	0.000	0.004*	0.005*	0.004	0.002
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (M4)	-0.002	0.000	0.004*	0.005*	0.004	0.002
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (M5)	-0.002	-0.001	0.003	0.005	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (M6)	-0.002	-0.000	0.003	0.005*	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
% Manufacturing (LMB)	-0.002	0.000	0.004*	0.005*	0.004	0.003
	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)	(0.004)
Bandwidth	All	+/− 25	+/− 10	+/− 5	+/− 2	Polynomial
Observations	13413	10229	4174	2072	810	13393

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Standard errors are in parenthesis. The unit of observation is the district-congress. Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: LMB's Balance Tests Using Extract Data Versus Our Harmonized Data (V)

	Difference in District Voters Between Democrat and Republican Districts					
	(1)	(2)	(3)	(4)	(5)	(6)
% Eligible to vote (M1)	0.004	0.008***	0.007**	0.006	-0.006	-0.013*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
% Eligible to vote (M2)	0.004	0.009***	0.007**	0.006	-0.006	-0.013*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
% Eligible to vote (M3)	0.004	0.009***	0.007**	0.006	-0.006	-0.012*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
% Eligible to vote (M4)	0.004	0.009***	0.007**	0.006	-0.006	-0.012*
	(0.003)	(0.003)	(0.003)	(0.004)	(0.006)	(0.007)
% Eligible to vote (M5)	0.005	0.008***	0.008**	0.006	-0.007	-0.011
	(0.003)	(0.003)	(0.004)	(0.004)	(0.006)	(0.007)
% Eligible to vote (M6)	0.005*	0.009***	0.008**	0.006	-0.007	-0.011
	(0.003)	(0.003)	(0.004)	(0.004)	(0.007)	(0.007)
% Eligible to vote (LMB)	0.005*	0.011***	0.007**	0.007*	-0.003	-0.004
	(0.003)	(0.003)	(0.003)	(0.004)	(0.006)	(0.006)
Bandwidth	All	$+/-$ 25	$+/-$ 10	$+/-$ 5	$+/-$ 2	Polynomial
Observations	13413	10229	4174	2072	810	13393

Note: Each row features estimates from a different harmonization model, except for row (7), which uses data and code from Lee et al. (2004). Observation counts reflect those in row (7). Standard errors are in parenthesis. The unit of observation is the district-congress. Column (1) features the entire sample. Columns (2) through (5) limit the sample by varying bandwidths around the 50 percent mark. Column (6) includes a fourth order polynomial in Democratic vote share, which is interacted with the above-below 50 percent dummy. Standard errors are clustered by district-decade. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

References

Bazzi, Samuel, Martin Fiszbein, and Mesay Gebresilasse, "Frontier Culture: The Roots and Persistence of "Rugged Individualism" in the United States," *Econometrica*, 2020, *88* (6), 2329–2368.

Eckert, Fabian, Andres Gvirtz, Jack Liang, and Michael Peters, "A Method to Construct Geographical Crosswalks with an Application to US Counties since 1790," *NBER Working Paper No. 26770*, 2020.

Fang, Yu and James W. Jawitz, "High-resolution reconstruction of the United States human population distribution, 1790 to 2010," *Scientific Data*, 2018, *5*, https://doi.org/10.1038/sdata.2018.67.

Haines, Michael, "Historical, Demographic, Economic, and Social Data: The United States, 1790-2002," *Inter-university Consortium for Political and Social Research [distributor], Ann Arbor, MI, 2010-05-21. https://doi.org/10.3886/ICPSR02896.v3*, 2010.

Hornbeck, Richard, "Barbed Wire: Property Rights and Agricultural Development," *Quarterly Journal of Economics*, 2010, *125* (2), 767–810.

Lee, David S., Enrico Moretti, and Matthew J. Butler, "Do Voters Affect or Elect Policies? Evidence from the U. S. House," *Quarterly Journal of Economics*, 2004, *119* (3), 807–859.

Lewis, Jeffrey B., Brandon DeVine, Lincoln Pritcher, and Kenneth C. Martis, *United States Congressional District Shapefiles*, 2021, *https://cdmaps.polisci.ucla.edu/ (Accessed on June 30, 2021).*

Leyk, Stefan, Johannes H. Uhl, Dylan S. Connor, Anna E. Braswell, Nathan Mietkiewicz, Jennifer K. Balch, and Myron Gutmann, "Two Centuries of Settlement and Urban Development in the United States," *Science Advances*, 2020, *6*, 1–12.

Manson, Steven, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles, "IPUMS National Historical Geographic Information System," *Version 15.0 [dataset]. Minneapolis, MN. DOI: http://doi.org/10.18128/D050.V15.0.*, 2020.

Perlman, Elisabeth, "Tools for Harmonizing County Boundaries," *http://elisabethperlman.net/code.html. Accessed on May 18, 2021*, 2021.

Uhl, Johannes H. and Stefano Leyk, "Historical built-up areas (BUA) - gridded surfaces for the U.S. from 1810 to 2015," *Harvard Dataverse*, 2020, *V1*, https://doi.org/10.7910/DVN/J6CYUJ.

_ and _ , "Historical built-up property records (BUPR) - gridded surfaces for the U.S. from 1810 to 2015," *Harvard Dataverse*, 2020, *V1*, https://doi.org/10.7910/DVN/YSWMDR.